

First year Aerospace Engineering

Mathematics Recap - Integrals

Marcos Carrera Valle

Simple Integrals - 6 points

Find the integrals of the following functions

1) $\int 4x^3 + 6x^2 + 12x + 4 + x^{-2} dx$

2) $\int \frac{1}{x} dx$

3) $\int e^x dx$

4) $\int \sin(x) dx$

5) $\int \cos(x) dx$

6) $\int 3^x dx$

Complex Integrals - 8 points

Solve the following

1) $\int 2xe^x dx$

2) $\int_{\frac{\pi}{2}}^{\frac{5\pi}{8}} \sin(2x - \pi) dx$

3) $\int \frac{1}{x^2 - 4} dx$

4) $\int \ln(4x + 1) dx$

Solids of revolution - 6 points

Calculate the surface area and volume of the solid of revolution formed by rotating the function $f(x) = \sqrt{x}$ by 360° around the x-axis from $x = 0$ to $x = 4$

Answers

Slide 2

1) $x^4 + 2x^3 + 6x^2 + 4x - x^{-1} + C$

2) $\ln(x) + C$

3) $e^x + C$

4) $-\cos(x)$

5) $\sin(x)$

6) $\frac{3^x}{\ln(3)} + C$

Answers

Slide 3

1) $2e^x(x-1)+C$

2) $\frac{\sqrt{2}-2}{4}$

3) $\frac{1}{4}\ln\left(\frac{x-2}{x+2}\right)+C$

4) $\left(x+\frac{1}{4}\right)(\ln(4x+1)-1)+C$

Answers

Slide 4

$$\text{Volume} = 8\pi$$

$$\text{Surface Area} = \frac{32\pi}{3}$$